

Assignment 1

Due: Tuesday, October 7, 2025 (before 10 am via eClass)

Question 1 (15 points):

In a few sentences, or with a brief proof, answer the following questions.

- (a) Define a Projection matrix.
- (b) What does the Gauss Markov Theorem say?
- (c) Derive the conditional variance of the least squares residual vector in the multiple linear regression.
- (d) Derive the conditional variance of the least squares estimator vector in the multiple linear regression.
- (e) Using Frisch-Waugh-Lovell Theorem, derive the conditional variance of a particular element of the least squares estimator vector in the multiple linear regression.

Question 2 (10 points):

Let the random variable X be the time in seconds between incoming telephone calls at a busy switchboard. Suppose that a reasonable probability density function for X is given by

$$f(x) = \begin{cases} \frac{1}{4} e^{-\frac{x}{4}} & \text{if } 0 < x < \infty \\ 0 & \text{elsewhere.} \end{cases}$$

1. Verify that $\int_{-\infty}^{\infty} f(x)dx = 1$ and find the CDF of X .
2. Compute the probability that the time between successive phone calls exceeds 4 seconds.

Question 3 (5 points):

It is known that $E(X|Y) = E(X)$ if X and Y are independent. Using this fact, show that $E(XY) = E(X)E(Y)$ if X and Y are independent.

Question 4 (10 points):

Consider the linear regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon} \sim IID(\mathbf{0}, \sigma^2 \mathbf{I}_n),$$

where $\mathbf{y}$ and $\boldsymbol{\varepsilon}$ are $n \times 1$, $\mathbf{X}$ and $\mathbf{Z}$ are $n \times k$, and $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ are $k \times 1$. Assume that $\mathbf{X}$ and $\mathbf{Z}$ are non-random. Let $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\gamma}})$ denote the OLS estimate from regressing $\mathbf{y}$ on $\mathbf{X}$ and $\mathbf{Z}$. Let $\tilde{\boldsymbol{\beta}}$ denote the OLS estimate from regressing $\mathbf{y}$ on $\mathbf{X}$ only, and let $\tilde{\boldsymbol{\gamma}}$ denote the OLS estimate from regressing $\mathbf{y}$ on $\mathbf{Z}$ only. Suppose $\mathbf{X}'\mathbf{Z} = \mathbf{0}$. Show that

1. $\hat{\boldsymbol{\beta}} = \tilde{\boldsymbol{\beta}}$,
2. $\hat{\boldsymbol{\gamma}} = \tilde{\boldsymbol{\gamma}}$,
3. $E((\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})(\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma})') = \mathbf{0}$.

Question 5 (10 points):

Suppose that $\mathbf{y}$ is a $n \times 1$ vector, $\mathbf{X}$ is a $n \times k$ matrix, and $\mathbf{Z} = \mathbf{X}\mathbf{B}$ where, $\mathbf{B}$ is a $k \times k$ invertible matrix. Denote $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\varepsilon}}$ the OLS estimator and the vector of residuals regressing $\mathbf{y}$ on $\mathbf{X}$. Denote also $\tilde{\boldsymbol{\beta}}$ and $\tilde{\boldsymbol{\varepsilon}}$ the OLS estimator and the vector of residuals regressing $\mathbf{y}$ on $\mathbf{Z}$. What is the relation between $\hat{\boldsymbol{\beta}}$ and $\tilde{\boldsymbol{\beta}}$, and between $\hat{\boldsymbol{\varepsilon}}$ and $\tilde{\boldsymbol{\varepsilon}}$.

Question 6 (20 points):

The file `ceosal2.dta` contains data on 177 chief executive officers and can be used to examine the effects of firm performance on CEO salary.

1. Estimate a model relating annual salary to firm sales and market value. Make the model of the constant elasticity variety for both independent variables. Write the results out in equation form.
2. Add profits to the model from part (1). Why can this variable not be included in logarithmic form? Would you say that these firm performance variables explain most of the variation in CEO salaries?
3. Add the variable `ceoten` to the model in part (2). What is the estimated percentage return for another year of CEO tenure, holding other factors fixed?

Question 7 (30 points):

1. The file tbrate.xlsx contains data for 1950 :1 to 1996 :4 for three series: r_t , the interest rate on 90-day treasury bills, π_t , the rate of inflation, and y_t , the logarithm of real GDP.

1.1. For the period 1950 :4 to 1996 :4, run the regression

$$\Delta r_t = \beta_1 + \beta_2 \pi_{t-1} + \beta_3 \Delta y_{t-1} + \beta_4 \Delta r_{t-1} + \beta_5 \Delta r_{t-2} + u_t \quad (1)$$

where Δ is the first-difference operator, defined so that $\Delta r_t = r_t - r_{t-1}$.

1.2. Plot the residuals and fitted values against time.

1.3. Regress the residual on the fitted values and on a constant. What do learn from this second regression.

1.4. Now regress the fitted values on the residuals and on a constant. What do learn from this third regression ?

2. For the same sample, regress Δr_t on a constant, Δy_{t-1} , Δr_{t-1} and Δr_{t-2} . Save the residuals from this regression, and call them $\hat{e}_t$. Then regress π_{t-1} on a constant, Δy_{t-1} , Δr_{t-1} and Δr_{t-2} . Save the residuals from this regression, and call them $\hat{v}_t$. Now regress $\hat{e}_t$ on $\hat{v}_t$.
3. How does the estimated coefficient and the residuals from this last regression related to anything that you obtained when you estimated regression (1).